CrimeWatch – Crime Hotspot Detection using Machine Learning

ABSTRACT

This project titled **"CrimeWatch – Crime Hotspot Detection using Machine Learning"** focuses on analyzing crime data to identify crime-prone locations using unsupervised learning techniques. The system uses clustering algorithms to group crime incidents based on geographical coordinates such as latitude and longitude.

The project is implemented using **Python, OpenCV libraries, Scikit-Learn, and Streamlit**, providing an interactive web-based application for visualizing crime clusters on a map. The system allows users to upload a crime dataset, process the data using clustering algorithms, and visualize the detected crime hotspots.

The application also provides visualizations such as cluster plots and an interactive map showing crime locations. This helps in identifying patterns in crime occurrences and supports decision-making for law enforcement agencies, urban planners, and researchers.

INTRODUCTION

Crime analysis plays an important role in improving public safety and understanding patterns in criminal activities. Traditional crime monitoring methods rely on manual analysis of reports and statistics, which can be time-consuming and inefficient.

Machine learning provides powerful tools to analyze large datasets and discover hidden patterns. **Unsupervised learning techniques such as clustering** can automatically group similar data points and identify regions where crimes occur frequently.

This project uses **clustering algorithms to detect crime hotspots** by analyzing crime location data. The system uses **Principal Component Analysis (PCA)** for dimensionality reduction and **DBSCAN clustering** to group nearby crime incidents.

A **Streamlit web application** provides an interactive interface where users can upload crime data, view clusters, and visualize crime hotspots on a map.

OBJECTIVES

The main objectives of this project are:

- To analyze crime data using machine learning techniques
- To detect crime hotspots using clustering algorithms
- To visualize crime clusters using graphical plots
- To display crime locations on an interactive map
- To allow users to upload crime datasets easily
- To provide an option to download clustered results
- To build a simple and user-friendly web interface using Streamlit

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SYSTEM REQUIREMENTS

Hardware Requirements

• Computer or Laptop
• Minimum 4 GB RAM
• Internet connection (optional for datasets)

Software Requirements

• Operating System: Windows / Linux / MacOS
• Python 3.8 or above
• Streamlit
• Scikit-Learn
• Pandas
• NumPy
• Folium
• Plotly

TECHNOLOGIES USED

Python

Python is used as the main programming language due to its simplicity and strong support for data science and machine learning libraries.

Scikit-Learn

Scikit-Learn is used for implementing machine learning algorithms such as:

• DBSCAN clustering
• PCA (Principal Component Analysis)

Streamlit

Streamlit is used to build the web-based interface of the system. It allows:

• Uploading datasets
• Displaying graphs and visualizations
• Showing interactive maps
• Downloading processed data

Folium

Folium is used to generate interactive maps that display crime locations and clusters.

Plotly

Plotly is used for creating interactive visualizations such as cluster scatter plots.

NumPy

NumPy is used for efficient numerical computations and data processing.

MODULE DESCRIPTION

Data Upload Module

- Allows users to upload a CSV dataset
- Reads and displays dataset preview
- Checks required columns such as latitude and longitude

Data Processing Module

- Extracts coordinates from the dataset
- Normalizes the data using StandardScaler
- Applies PCA for dimensionality reduction

Clustering Module

- Applies DBSCAN clustering algorithm
- Groups crime incidents into clusters
- Identifies noise points or isolated crimes

Visualization Module

- Generates cluster visualization using Plotly
- Displays crime clusters in a 2D scatter plot

Map Visualization Module

- Displays crime locations on an interactive map
- Marks clusters using different colors
- Shows city and crime type information on click

Download Module

- Allows users to download the dataset with cluster labels
- Saves results in CSV format

WORKING PRINCIPLE

1. The user uploads a crime dataset in CSV format.
2. The system reads and displays the dataset preview.
3. Latitude and longitude values are extracted.
4. Data is normalized using StandardScaler.
5. PCA reduces dimensions for visualization.
6. DBSCAN clustering groups nearby crime incidents.
7. Cluster results are displayed using graphs.
8. Crime locations are plotted on an interactive map.
9. Users can download the clustered dataset.

ALGORITHM

1. Start the application
2. Upload crime dataset
3. Extract Latitude and Longitude values
4. Normalize data using StandardScaler
5. Apply PCA for dimensionality reduction
6. Apply DBSCAN clustering algorithm
7. Assign cluster labels to each crime record
8. Visualize clusters using Plotly
9. Display crime locations on Folium map
10. Allow user to download clustered dataset

ADVANTAGES

• Automatically identifies crime hotspots
• Easy-to-use web interface
• Supports real-time visualization
• Helps in analyzing crime patterns
• Useful for law enforcement agencies
• Supports large datasets

APPLICATIONS

- Crime analysis for police departments
- Urban planning and safety monitoring
- Smart city surveillance systems
- Research in criminology and public safety
- Academic machine learning experiments

LIMITATIONS

- Requires accurate location data
- Dataset quality affects clustering results
- Large datasets may slow processing
- Depends on proper parameter tuning

FUTURE ENHANCEMENTS

- Real-time crime data integration
- Advanced clustering algorithms
- Crime prediction using supervised learning
- Dashboard analytics for crime trends
- Integration with GIS systems
- Mobile application support

CONCLUSION

The **CrimeWatch – Crime Hotspot Detection System** demonstrates how machine learning techniques can be used to analyze crime data effectively. By using clustering algorithms such as DBSCAN and dimensionality reduction through PCA, the system identifies crime hotspots and visualizes them on an interactive map.

The integration of **Python, Streamlit, Scikit-Learn, and visualization tools** creates a simple and efficient platform for crime data analysis. This project highlights the practical application of unsupervised learning in real-world scenarios and can be further extended for advanced crime analytics and smart city safety systems.